

Mitigating Physical Layer Vulnerabilities in 6G Communications using Deep Reinforcement Learning

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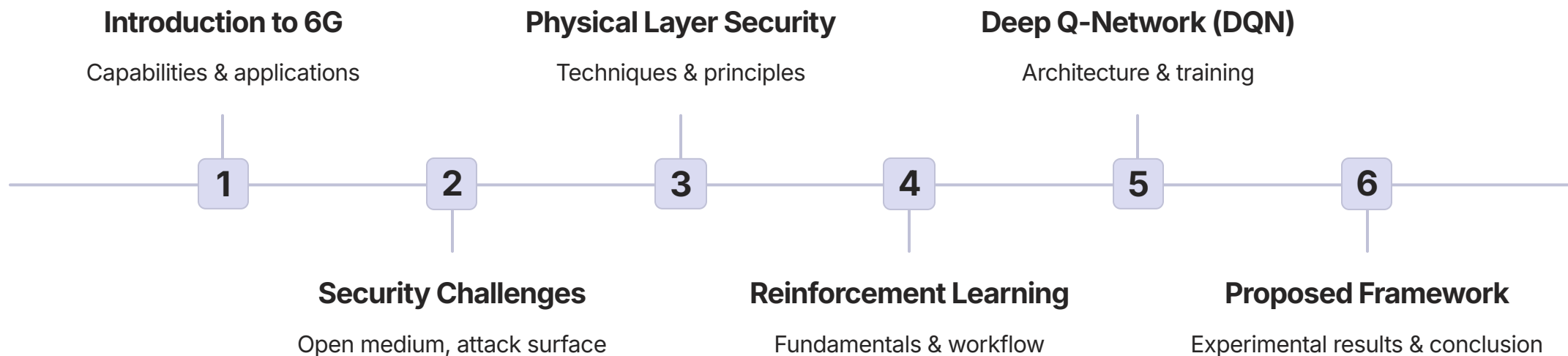
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Presentation Roadmap

This presentation introduces the 6G landscape, explores its unique security vulnerabilities, and builds toward a novel AI-driven defense framework. Follow the journey from fundamentals to the proposed solution.





What is 6G?

6G is the **sixth generation of wireless communication systems**, expected to be deployed around 2030. It goes far beyond incremental speed improvements — 6G is envisioned as an AI-native, fully adaptive communication fabric that underpins intelligent autonomous systems at a global scale.

Expected Capabilities

- Ultra-high data rates (Tbps range)
- Ultra-low latency (<1 ms)
- Massive device connectivity
- AI-native networking
- Intelligent autonomous systems

Key Application Domains

- Smart Cities & infrastructure
- Autonomous Vehicles
- Industrial Automation
- Massive Internet of Things (IoT)

Key Technologies Enabling 6G

Several breakthrough technologies converge to make 6G possible. Each addresses a specific bottleneck in capacity, coverage, or intelligence — together, they form the technical foundation of next-generation wireless networks.



Terahertz Communication

Operates in the THz frequency band, enabling data rates in the terabit-per-second range for ultra-fast short-range links.



Artificial Intelligence

AI is embedded natively into the network architecture, enabling real-time resource management, anomaly detection, and self-optimization.



Massive MIMO

Deploys hundreds of antennas at base stations to maximize spatial multiplexing, improving coverage and spectral efficiency.



Intelligent Networking

Supports adaptive, context-aware communication protocols that can autonomously reconfigure based on environmental conditions.

Security Challenges in 6G

While 6G delivers extraordinary performance gains, its very design introduces an expanded and more complex security threat landscape. The openness of the wireless medium, combined with the scale and heterogeneity of 6G deployments, makes it a prime target for sophisticated adversaries.

Open Wireless Medium

Unlike wired networks, wireless signals propagate freely — any receiver within range can potentially intercept transmissions.

Dynamic Environments

Rapidly changing channel conditions and mobile topologies make it difficult to maintain consistent security policies.

Massive Attack Surface

Billions of connected IoT devices and heterogeneous nodes create countless entry points for adversarial exploitation.

Intelligent Attackers

AI-enabled adversaries can adapt their strategies in real time, making static rule-based defenses increasingly ineffective.

❓ **Core Question:** How can wireless communication remain secure and reliable in such a dynamic and hostile environment?

Physical Layer Vulnerabilities

Wireless signals are inherently exposed to the physical environment. Because no physical boundary exists between transmitter and receiver, the communication medium itself becomes a vulnerability. Attackers who understand channel characteristics can exploit this openness in several powerful ways.

Nature of the Threat

Unlike encrypted data at higher layers, raw wireless signals travel through open air — they can be captured, disrupted, or impersonated before any higher-layer protection can act.

- Signals can be intercepted passively
- Signals can be actively disrupted
- Attackers can exploit channel characteristics for deception

Common Physical Layer Threats

- **Jamming** — Deliberate interference to block communication
- **Eavesdropping** — Passive interception of transmitted data
- **Spoofing** — Impersonation of legitimate nodes to inject false data

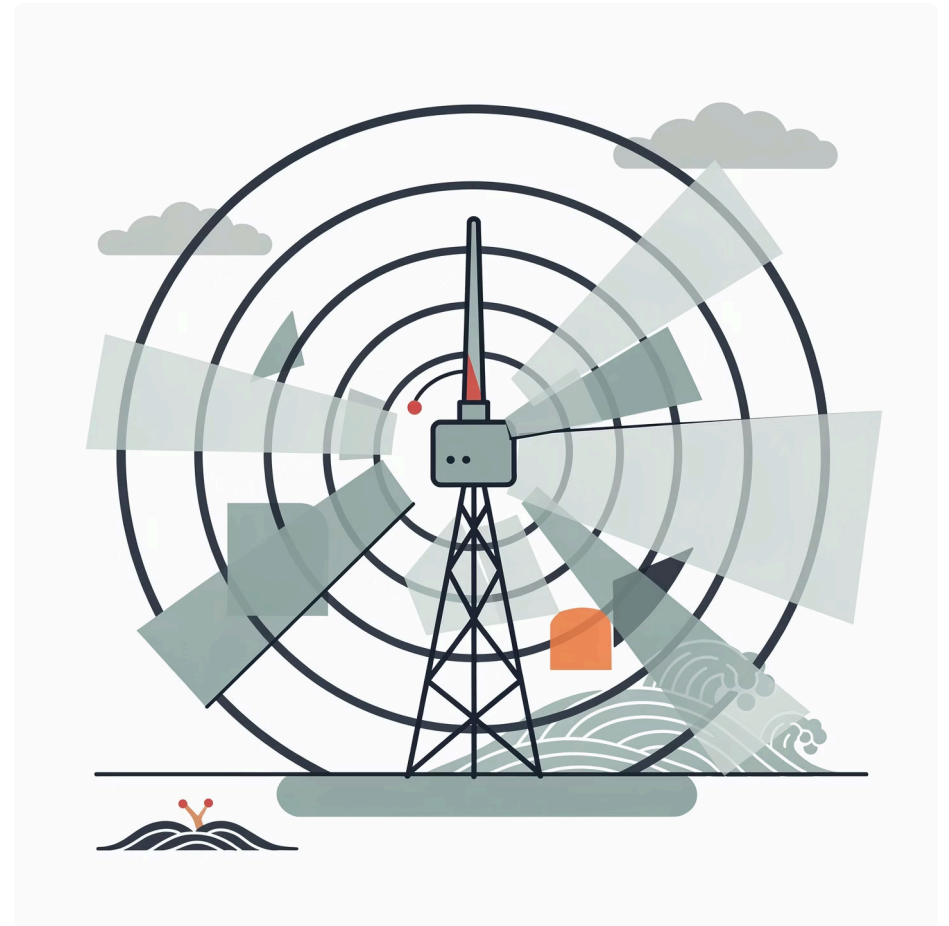
Jamming Attacks

A **jamming attack** occurs when a malicious node intentionally transmits interference signals on the same frequency band as the target communication channel. The jammer does not need to decode any data — its sole purpose is disruption. Jamming can be constant, intermittent, or reactive, making it especially difficult to counter with static defenses.

Effects on the Network

- Degraded signal-to-noise ratio (SNR)
- Increased packet error rates
- Communication interruptions and outages
- Reduced reliability and throughput

⊗ Attacker Goal: Deny legitimate users access to the communication channel — a form of Denial-of-Service (DoS) at the physical layer.



Eavesdropping Attacks



In an **eavesdropping attack**, an unauthorized entity passively monitors wireless transmissions without the knowledge of communicating parties. Because no active interference is introduced, eavesdropping is extremely difficult to detect using conventional monitoring techniques.

Consequences

- Sensitive data leakage (credentials, commands, telemetry)
- Serious privacy violations for end users
- Loss of communication confidentiality
- Enables follow-on attacks using captured information

⚠ **Attacker Goal:** Extract sensitive information from wireless transmissions without being detected — a passive but highly damaging threat.



Traditional Security Approaches

Conventional communication security relies on a well-established stack of cryptographic and access control mechanisms. These techniques have proven effective for decades across internet and cellular systems.

Encryption

Transforms data into unreadable ciphertext using cryptographic keys. Protects data confidentiality in transit and at rest.

Authentication

Verifies the identity of communicating parties using certificates, tokens, or challenge-response protocols.

Access Control

Enforces policies that restrict which users and devices may access network resources and services.

- ❑ These techniques operate primarily at the **application, transport, and network layers** — well above the physical wireless medium where many 6G threats originate.

Why Traditional Security Is Not Enough

While encryption and authentication remain essential, they were not designed to address threats that operate directly at the wireless transmission medium. In the context of 6G — with billions of low-power devices and sub-millisecond latency requirements — their limitations become critical bottlenecks.

1

Computational Overhead

Cryptographic operations require processing power that many resource-constrained IoT devices simply cannot afford.

2

Additional Latency

Key exchange and handshake protocols introduce delays incompatible with 6G's ultra-low latency requirements.

3

Scalability Limitations

Managing cryptographic identities for billions of 6G-connected devices poses enormous key management challenges.

4

Physical Layer Blind Spot

Encryption cannot stop a jamming signal or prevent eavesdropping at the radio frequency level — the attack happens before decryption is even possible.

What is Physical Layer Security?

Core Concept

Physical Layer Security (PLS) leverages the inherent properties of the wireless channel — such as fading, noise, and spatial diversity — to provide information-theoretic security guarantees. Rather than adding encryption on top of transmission, PLS makes the channel itself a security mechanism.

Theoretical Foundation

PLS is grounded in Shannon's secrecy capacity and Wyner's wiretap channel model: secure communication is possible when the legitimate channel is statistically better than the eavesdropper's channel.

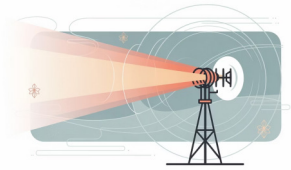
Key Benefits

- **Lightweight** — No heavy cryptographic computation required
- **Adaptive** — Responds dynamically to channel changes
- **Complementary** — Works alongside, not instead of, cryptography
- **Future-Ready** — Well-suited for the scale and dynamism of 6G



Physical Layer Security Techniques

A range of PLS techniques have been developed to exploit wireless channel properties for defense. Each targets a different aspect of the physical transmission to degrade the attacker's ability to intercept or disrupt communication.



Beamforming

Concentrates transmitted signal energy in the precise direction of the legitimate receiver, minimizing signal leakage in other directions where attackers may reside.



Power Allocation

Dynamically adjusts the transmission power level based on real-time channel state information to maximize the secrecy capacity of the link.



Artificial Noise Injection

Injects controlled noise into the null space of the legitimate channel, degrading the eavesdropper's received signal quality without affecting the intended receiver.



Channel-Based Authentication

Uses unique channel fingerprints — multipath fading patterns specific to each link — to verify the identity of communicating nodes without cryptographic keys.

Artificial Intelligence in Wireless Networks

The complexity and dynamism of 6G networks make manual optimization impossible. **Artificial Intelligence** provides the adaptive intelligence needed to manage these systems in real time — learning from the environment and making decisions that no hand-crafted algorithm could match.

Pattern Learning AI models identify regularities in channel behavior, traffic patterns, and attack signatures — enabling proactive rather than reactive responses.	Adaptive Optimization Continuously adjusts network parameters — power, frequency, beamforming vectors — in response to changing conditions and emerging threats.	Autonomous Decisions Enables self-configuring networks that minimize human intervention while maintaining security and performance guarantees.
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Key AI application areas in wireless security include **resource allocation**, **threat detection**, **security policy optimization**, and **intelligent network management**.

Reinforcement Learning Fundamentals

Reinforcement Learning (RL) is a machine learning paradigm in which an agent learns optimal behavior by interacting with an environment through trial and error. Unlike supervised learning, RL requires no labeled dataset — the agent discovers strategies by maximizing cumulative reward signals over time. This makes RL ideal for dynamic wireless security problems where optimal policies cannot be pre-specified.



Agent

The **decision-maker** — in our context, the intelligent security controller embedded in the 6G base station or node.



Environment

The **wireless network** — including the channel conditions, active jammers, eavesdroppers, and legitimate users.



Action

A **decision taken by the agent** — e.g., adjusting transmission power, switching frequency band, or modifying beamforming direction.

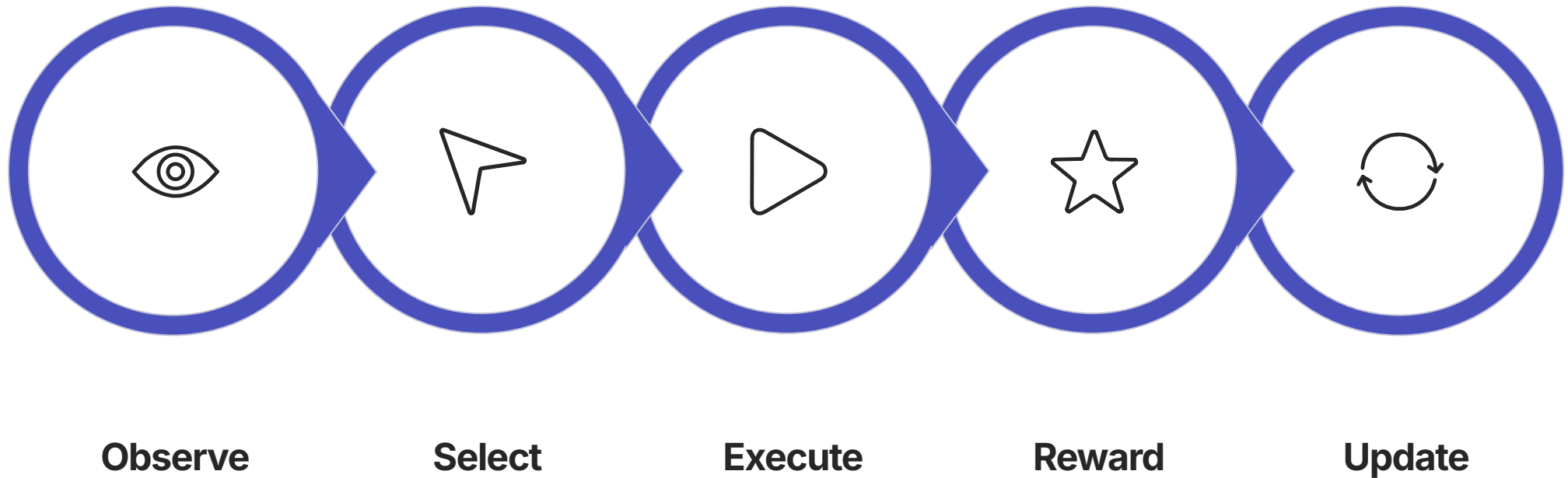


Reward

A **scalar feedback signal** received after each action — quantifying how well the chosen action improved security and communication quality.

Reinforcement Learning Workflow

The RL agent operates in a continuous learning cycle, refining its policy with every interaction. The objective is to find the optimal policy π^* — the mapping from states to actions that maximizes the expected cumulative discounted reward over time. In wireless security, this translates to consistently choosing actions that improve secrecy capacity while resisting attacks.



- ✓ **Convergence Goal:** Through repeated interaction, the agent's policy converges toward optimal decisions — maximizing cumulative reward and achieving robust, adaptive physical layer security without requiring explicit programming of every scenario.

Deep Reinforcement Learning (DRL)

MODULE 4 · ADVANCED METHODS

Deep Reinforcement Learning (DRL) represents a powerful convergence of two foundational paradigms in machine learning. By fusing classical **Reinforcement Learning** — where an agent learns by interacting with its environment — with the representational power of **Deep Neural Networks**, DRL can tackle problems that were previously intractable.

Core Combination

- Reinforcement Learning: trial-and-error decision making
- Deep Neural Networks: high-dimensional function approximation

Key Advantages

- Handles large, continuous state spaces
- Learns complex, non-linear policies
- Suitable for dynamic, unpredictable environments like 6G channels



Deep Q-Network (DQN)

CORE ALGORITHM

The proposed security framework is built around the **Deep Q-Network (DQN)** — one of the most influential DRL algorithms. DQN extends Q-Learning by replacing the traditional Q-table with a deep neural network, enabling it to approximate action-value functions in high-dimensional spaces. Its design choices directly address the instability challenges that arise when combining neural networks with reinforcement learning.



Neural Network

Approximates the Q-value function, mapping states to expected rewards for each possible action.



Q-Learning

Iteratively updates action-value estimates using the Bellman equation to converge on an optimal policy.



Experience Replay

Stores past transitions and samples mini-batches randomly, breaking temporal correlations for stable training.



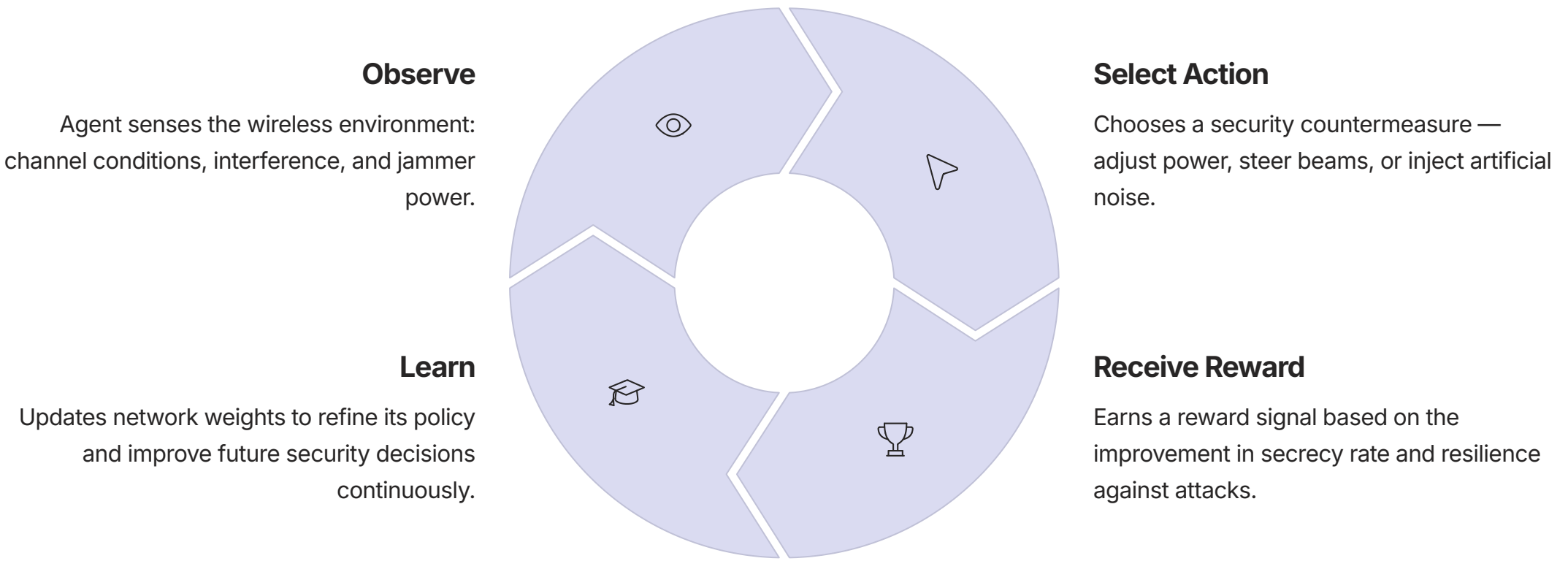
Epsilon-Greedy

Balances exploration of new actions with exploitation of known good strategies during training.

Proposed DQN Security Framework

SYSTEM OVERVIEW

The central goal of this research is to **protect 6G communications** against two critical threat categories: **jamming attacks** (which degrade signal quality) and **eavesdropping attacks** (which compromise confidentiality). The DQN agent operates as an intelligent, adaptive defender embedded within the communication system.



System Model & MDP Formulation

MATHEMATICAL FOUNDATION

The wireless security problem is formally modeled as a **Markov Decision Process (MDP)**. At each time step, the DQN agent perceives a state capturing the current channel conditions, selects an action to adapt the transmitter, and receives a reward. This formulation allows rigorous application of reinforcement learning theory.

State Space — S_t

The agent observes three key environmental variables:

- **CSI** — Channel State Information: characterizes the propagation path between transmitter and receiver
- **SINR** — Signal-to-Interference-plus-Noise Ratio: measures link quality under interference
- **P_j** — Jammer Power: estimated power level of the adversarial jammer

Action Space — A_t

The agent can adapt three transmitter parameters:

- **P_t** — Power Allocation: controls transmit power to balance capacity and interference
- **w_t** — Beamforming Control: steers the signal beam toward the legitimate receiver
- **N_t** — Artificial Noise Injection: adds deliberate noise to degrade the eavesdropper's channel

Experimental Setup

IMPLEMENTATION DETAILS

All experiments were conducted in a **Python-based simulation environment** designed to replicate dynamic 6G threat conditions. The DQN agent was trained under varying jamming and eavesdropping scenarios, ensuring it learned a robust, generalizable policy rather than overfitting to a single channel configuration.

Simulation Environment

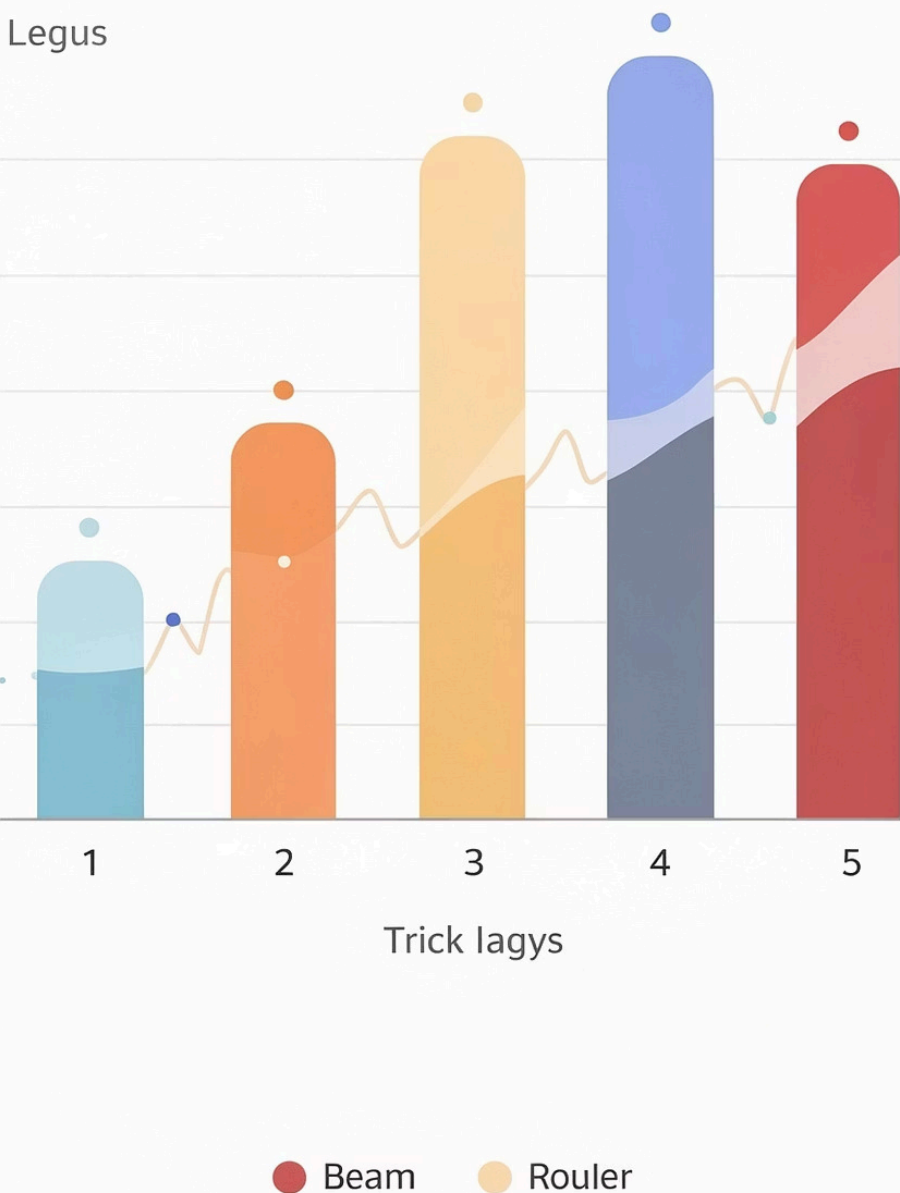
- Python simulation framework
- Trained DQN agent
- Dynamic jamming conditions (variable power)
- Dynamic eavesdropping conditions (varying positions)

Training Hyperparameters

Parameter	Value
Learning Rate	0.001
Discount Factor (γ)	0.99
Replay Buffer Size	10,000
Batch Size	32
Optimizer	Adam
Activation Function	ReLU

Performance Comparison

Legus



Experimental Results

KEY OUTCOME

The DQN-based security framework was benchmarked against a **static baseline** — a non-adaptive policy that applies fixed transmit parameters regardless of environmental changes. The results demonstrate a decisive advantage for the learned, adaptive approach.

3.84

DQN Reward

Average cumulative reward achieved by the trained DQN agent across all evaluation episodes.

2.57

Static Reward

Average cumulative reward of the non-adaptive static baseline under identical conditions.

49.28%

Improvement

Overall performance gain of the DQN framework over the static method — a statistically significant result.

- ✓ The DQN framework significantly outperformed the static baseline, validating that adaptive, learning-based security strategies are superior for dynamic 6G threat environments.

Performance Comparison

DETAILED ANALYSIS

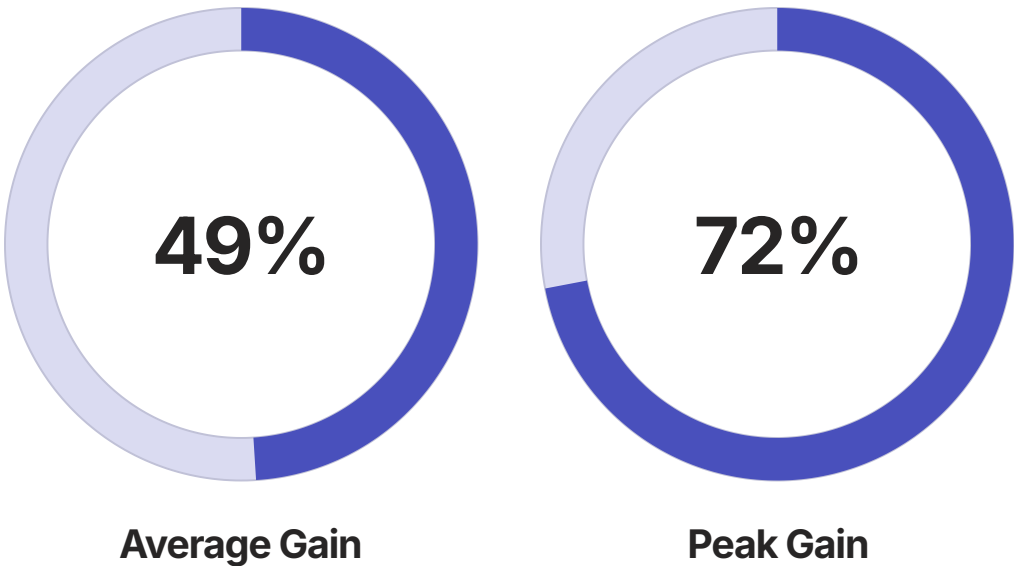
A closer look at individual evaluation runs reveals the **consistency and strength** of the DQN agent's advantage. While the aggregate improvement stands at 49.28%, individual episodes highlight even more dramatic gains — demonstrating the agent's ability to capitalize on favorable channel conditions.

Key Observations

- DQN outperformed the static method in the **vast majority** of evaluation runs
- Peak improvement in a single run reached **72.42%** — illustrating the agent's ceiling
- Only **one run** showed lower DQN performance, indicating a rare edge case
- The **49.28% average improvement** is a robust, aggregated measure of superiority

i The single underperforming run may reflect an environment state outside the agent's training distribution — a known challenge in RL generalization.

Performance Breakdown



Current Limitations

IMPORTANT CAVEATS

Scientific rigor demands transparency about the boundaries of this study. The current implementation is a **proof-of-concept prototype** designed to validate the DQN approach in a controlled setting. Several simplifications were intentionally made to enable focused experimentation, and these must be considered when interpreting the results.

Simplified Wireless Environment

The simulation uses idealized channel models that do not fully capture real-world 6G propagation effects such as mmWave scattering, mobility, or multipath fading.

Single-Agent Architecture

Only one DQN agent operates in the environment. Real deployments involve multiple base stations and users requiring coordinated, multi-agent decision-making.

Limited Attack Scenarios

The framework was evaluated against specific jamming and eavesdropping models. More sophisticated, adaptive adversaries were not considered.

Not Full-Scale 6G Deployment

Results cannot be directly extrapolated to production 6G networks without further validation in more realistic simulation or hardware testbed environments.

Future Research Directions

LOOKING AHEAD

The promising results from this study open several compelling avenues for future investigation. Each extension addresses one or more of the current limitations, progressively bridging the gap between this proof-of-concept and production-ready 6G Physical Layer Security systems.



Multi-Agent Reinforcement Learning

Extend the framework to cooperative and competitive multi-agent settings, enabling multiple base stations to jointly optimize security across a network.



RIS-Assisted Security

Integrate Reconfigurable Intelligent Surfaces (RIS) to dynamically shape the wireless channel, adding a powerful new dimension to physical-layer defense.



Realistic 6G Channel Models

Replace simplified models with standardized 3GPP 6G channel models including mmWave, THz bands, and high-mobility scenarios for more credible evaluation.



Advanced DRL Algorithms

Explore Proximal Policy Optimization (PPO), Soft Actor-Critic (SAC), and other state-of-the-art DRL algorithms to potentially surpass DQN performance.

Conclusion

KEY TAKEAWAYS

This research demonstrates that **AI-driven Physical Layer Security** is a viable and highly effective approach for protecting next-generation 6G wireless networks. The DQN-based framework successfully learned adaptive defense strategies in dynamic threat environments, outperforming conventional static methods by a substantial margin.

→ **Physical Layer Security is Essential**

As 6G networks scale in complexity, cryptographic approaches alone are insufficient — security must be embedded at the physical layer.

→ **Validated Performance: +49.28% Improvement**

The DQN framework achieved a 49.28% average improvement over the static baseline, with a peak gain of 72.42% in the best individual run.

→ **Reinforcement Learning Enables Adaptive Defense**

Unlike static policies, the DQN agent continuously adapts to changing attack conditions, making it inherently more resilient.

→ **AI-Driven Security is the Future**

This work establishes a strong foundation for intelligent, self-optimizing security in 6G — a direction that warrants continued investment and research.

Thank You. Questions are welcome — we look forward to discussing the DQN framework, MDP formulation, experimental methodology, and future research directions.